

nullCAT-Pi - GPIO Control Panel

Wiring & pin reference

One Cat5e/Cat6 cable carries every signal: 7 of 8 conductors used.

RJ45 / keystone breakout at each end. NOT a network link - do not plug into a NIC.

3.3 V logic only. Buttons use internal pull-ups; LEDs are sourced by the GPIO pins.

Conductor map

Function	BCM	Phys	Cat5e colour	Notes
E-STOP signal	GPI017	11	white-orange	NC mushroom to GND; open=stop
Common GND	GND	14	orange	shared return (one common wire)
ENGAGE	GPI027	13	white-green	momentary to GND
PARK	GPI022	15	green	momentary to GND
LED Run (grn)	GPI023	16	white-blue	pin -> 330R -> LED -> GND
LED Ready(amb)	GPI024	18	blue	pin -> 330R -> LED -> GND
LED Fault(red)	GPI025	22	white-brown	pin -> 330R -> LED -> GND
spare	-	-	brown	future 4th LED/button or 2nd GND

Panel behaviour

ENGAGE two-stage: offline -> Initialize EtherCAT ; ready (OP, stopped) -> Start loop

PARK running -> Park (returns platform to neutral; loop keeps running)

E-STOP open (or cable cut) -> disable drives. Release the mushroom to clear the e-stop and the UI (auto-release).

Status LEDs

-  **Green** solid = control loop running
-  **Amber** solid = ready (OP, stopped) ; blink = initializing
-  **Red** solid = E-STOP ; blink = drive fault
-  **All off** offline (EtherCAT not operational)

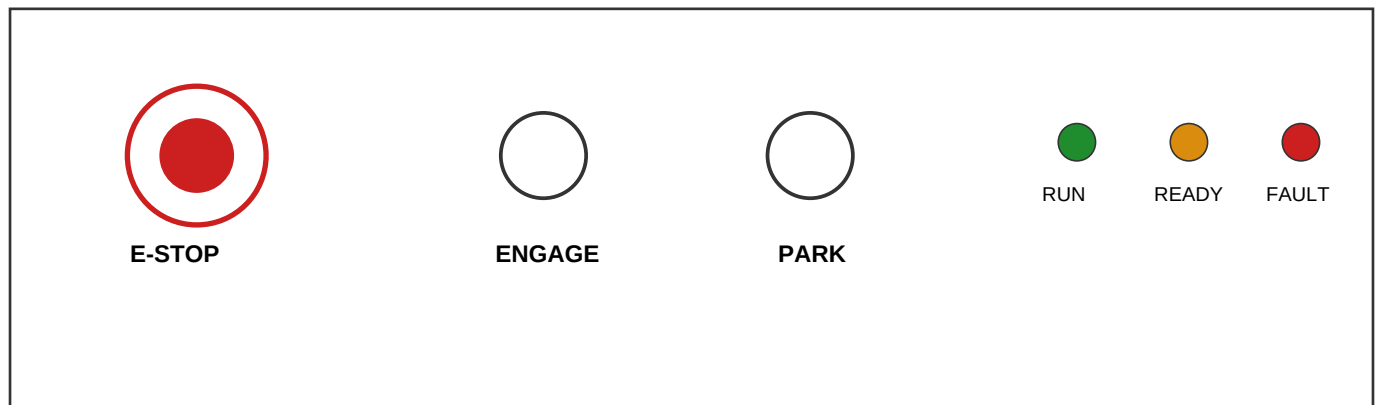
Pi 40-pin header - block used (pins 11-22)

Odd pins on the top rail, even on the bottom (Pi 4, gpiochip0).

pin 11 GPIO17 E-STOP	pin 13 GPIO27 ENGAGE	pin 15 GPIO22 PARK	pin 17 3V3 do NOT use	pin 19 GPIO10 unused	pin 21 GPIO9 unused
pin 12 GPIO18 unused	pin 14 GND COMMON	pin 16 GPIO23 LED Run	pin 18 GPIO24 LED Ready	pin 20 GND spare GND	pin 22 GPIO25 LED Fault

Tip: a 2x6 IDC/keyed block over pins 11-22 grabs all 7 conductors at once (leave pin 17 = 3V3 unwired).

Panel face layout (example)



Connectors & notes

Firmer than loose Dupont jumpers (in order): (1) screw-terminal GPIO HAT - wire each conductor into a labelled terminal, clamps the whole header; (2) 40-pin IDC ribbon + cobbler breakout; (3) a single 2x6 keyed crimp housing over pins 11-22.

E-STOP is NC + fail-safe: closed=OK (low), open OR a cut/unplugged cable = stop (high).

No separate power rail: buttons pull to the common GND; LEDs are driven by the GPIO pins.

Pins are config-driven (gpio* keys in config.json) - reassign freely and tell the firmware.

Enable in config.json: "gpioEnabled": true (build needs libgpiod-dev; user in 'gpio' group).